FEDERAL SPECIFICATION

BLOCK, TACKLE, WIRE ROPE, WITH SWIVEL HOOK; (FOR CRANE USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single and multiple sheave wire rope hook blocks with swivel hook for use with locomotive, crawler-, truck-, and wheel-mounted cranes.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all types, classes and styles of the commodity indicated by the title of the specification, but include only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Sizes.* Blocks shall be one of the following sizes, and capacities, as specified (see 6.1):

Size Capacity.
5.—5-ton
10.—10-ton
20.—20-ton
25.—25-ton
35.—35-ton
40.—40-ton
55.—55-ton
70.—70-ton
75.—75-ton

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:
MIL–S–890—Steel; Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL–F–3541—Fittings, Lubrication.
MIL–P–3865—Packaging and Packing for Overseas Shipment of Blocks (Wire and Manila Rope).

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–271—Nondestructive Testing Requirements for Metals.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Each block shall consist of one or more steel sheaves, steel side and separator plates, swivel hook, bearings, becket, and all other parts necessary for satisfactory operation. Adequate keepers shall be provided to prevent rope from leaving sheaves during operation. The block shall be new and unused.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of good commercial quality entirely suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.3 Interchangeability. Replaceable parts shall be manufactured to standard tolerances to permit replacement without requiring modification.

3.4 Design. The block shall conform to table I. The design shall be based on a factor of safety of not less than 5. The block shall be capable of withstanding the test loads without breakage or permanent distortion of any component part and shall fall free without load.

3.5 Sheaves. Sheaves shall be of steel and shall be designed for use with the wire-rope size specified. The outside diameter of the sheave shall be not less than 18 times the diameter of the wire rope for which the sheave is designed. Sheave groove shall be designed to give the rope a 150-degree arc of support with the sides of the groove tangent to the ends of the bottom arc. The depth of the groove shall be not less than 1.5 rope diameters. Sheave groove shall be machined true with the sheave bore and shall be hardened to a minimum of Rockwell C-35. Sheave hubs shall be machined and designed to bear against the side and separator plates with a smooth action.

3.6 Swivel hook. The swivel hook shall be forged steel. For Navy use, the steel shall conform to any class of MIL–S–890 with a longitudinal elongation in 2 inches of not less than 20 percent. Unless otherwise specified, the hook shall be the safety type fitted with a safety latch actuated by a stainless-steel spring. The hook shall be suspended from the block by means of a yoke or crosshead and shall swivel freely through 360 degrees when tested as specified in 4.6.4. After manufacturing processes

TABLE I. *Design Characteristics*

Capacity (safe working load in 2000-lb. tons)	Proof test load (2000-lb. tons)	Number of sheaves	Overall length, not more than (in.)	Gross weight, not less than (lbs.)	Hook opening, not less than (in.)
5	10	1	31	100	2
10	20	2	36	200	2
20	40	2	42	430	3
25	50	3	42	500	3
35	70	3	48	575	3–1/4
40	80	3	56	900	3–1/2
55	110	3	63	1000	4
70	140	3	64	1250	4
75	150	4	65	1300	4

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have been completed and prior to assembly, the hook shall show no cracks or other discontinuities when subjected to the magnetic particle test specified in 4.6.2. The narrowest part of the hook opening shall be not less than that shown in table I. When specified, the material designation and heat treatment shall be marked on the hook using combinations of the following applicable symbols and appropriate numbers: SAE (Society of Automotive Engineers), AISI (American Iron and Steel Institute), TS (tensile strength), H (heat with degrees given), N (normalized), A (anneal), OQ (oil quench), WQ (water quench), AQ (air quench), RH (reheat with degrees given), FS (furnace cool), AC (air cool).

3.7 Bearings.

3.7.1 *Sheave bearings.* Each sheave shall be mounted on one or more ball or roller bearings. Bearings shall be adequately sealed for retention of lubricant and exclusion of foreign matter.

3.7.2 *Swivel hook bearings.* Hook shall swivel on one or more ball or roller bearings. Bearings shall be of such size and quantity as to permit swiveling of capacity load without twisting or opening up the wire rope when tested as specified in 4.6.4. The bearings shall be adequately sealed for retention of lubricants and exclusion of foreign matter.

3.7.3 *Yoke or crosshead bearings.* When yoke or crosshead is designed to permit lateral movement of the hook, it shall be mounted in self-lubricating type porous phosphor bronze bushings.

3.8 Becket. Each block shall be supplied with a suitable becket. Becket shall show no evidence of breakage or permanent distortion when tested as specified in 4.6.1.

3.9 Cheek - plate weights. Cheek - plate weights, when utilized, shall be of steel. Plates shall be designed with recesses for boltheads and nuts.

3.10 Lubrication. Sheave and swivel hook bearings shall be provided with suitable means of lubrication. Hydraulic lubrication fittings shall conform to MIL–F–3541 and shall be located in accessible, protected positions. Each sheave bearing shall have an independent lubricating line and the line shall be located within the sheave pin.

3.11 Treatment and painting. All metal surfaces except bearing surfaces and hook shall be cleaned, treated, and painted in accordance with the manufacturer's best standard practice. The color shall be in accordance with Federal Standard No. 595, No. 13538, yellow gloss.

3.12 Identification markings. Each block shall be marked with the following information:

(*a*) Manufacturer's name or trademark.
(*b*) Manufacturer's part number.
(*c*) Safe working load in tons.
(*d*) Block size (i.e. sheave diameter).

The marking shall be applied to an identification plate (nameplate) which is securely fastened to the block, or may be applied directly to the surface of the block by acid or electric etching, engraving, embossing, casting, or molding. Marking shall be on the side of the block and shall be of such size and type as to be clearly legible after painting.

3.13 Workmanship.

3.13.1 *Steel fabrication.* Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.13.2 *Bolted connections.* Boltholes shall be accurately punched or drilled and shall

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have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.13.3 *Riveted connections.* Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neately made, concentric with the rivet holes, and shall be in full contact with the surface of the member.

3.13.4 *Welding.* The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by either the American Welding Society or the American Society of Mechanical Engineers for the type of welding operation to be performed.

3.13.5 *Machine work.* Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings and otherwise to the standards of good commercial practice.

3.13.6 *Castings.* Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for use.

3.13.7 *Forgings.* All forgings shall be of uniform quality, free of detrimental flash, scale, cracks, hard spots, cold shuts, and excessive cooling stresses. Strength and other essential physical properties of the forgings shall be adequate throughout for the purpose intended.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government for the period of time specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection requirements. The inspection requirements to be performed by the supplier are classified as follows:

(*a*) Production unit inspection.
(*b*) Preparation for delivery inspection.

4.3 Production unit inspection. All blocks produced in fulfillment of a Government contract or order shall be examined as specified in 4.4 and tested as specified in 4.6.1. and 4.6.2. Tests specified in 4.6.3 and 4.6.4. shall be conducted under the sampling procedure specified in 4.5.

4.4 Examination. Prior to submission to the Government for acceptance, each production unit shall be subjected to a visual, dimensional, or tactile examination for defects. Examinations shall be conducted during fabrication, manufacture, assembly, or after completion, as is manifestly appropriate or opportune. The following list of potential defects is provided for guidance and use. The list shall not be considered or interpreted to be all-inclusive or final. The Government reserves the right to include or add any other defects that arise or are found during the inspection processes, or that otherwise develop at any time prior to Government acceptance. Any production unit found to have one or more defects shall be rejected.

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LIST OF POTENTIAL DEFECTS

Examine	Requirement paragraphs	Defects
Material	3.2	Material used in fabrication or manufacture of any component or part not as specified. Obviously inferior or not suited to the purpose.
Design	3.4	Any applicable dimension or capacity not as specified in table I.
Sheaves	3.5	One or more dimensional characteristics not as specified. Hardness of groove does not meet minimum requirements.
Swivel hook	3.6	Any unauthorized deviation from or circumvention of one or more specified details.
Bearings–	3.7 through 3.7.3	Not in accordance with specified requirements.
Becket	3.8	Missing or inadequate.
Cheek weights	3.9	Not recessed for boltheads and nuts.
Lubrication	3.10	Means for lubrication not provided in one or more locations where lubrication is required or essential. One or more fittings missing, damaged, or not according to referenced specification. Provisions for sheave bearing lubrication not as specified.
Treatment and painting	3.11	Color of finish not as specified. Finish blistered, peeled, chipped, or otherwise impaired.
Identification markings	3.12	Missing or not affixed in prescribed manner, or insecurely fastened. Data incomplete, illegible, or erroneous.
Bolted and riveted connections	3.13.2	One or more missing, loose, deformed, or of a size not suited to the purpose.
Welding	3.13.4	Poor appearance, incomplete, porous, sparse, flux inclusions, excessive spatter, undercuts, or lack of fusion with parent metal or between parts.
Machine work	3.13.5	Surfaces rough, pitted, uneven, or excessively tool marked. Any machined surface condition adversely affecting tolerances or fit-ups of mating or working parts.
Castings	3.13.6	Blowholes, patches, warps, extraneous fins, or any other similar or characteristic defect impairing the function, operability, reliability or appearance of the piece, part, component, or assembly.
Forgings	3.13.7	Presence of cracks, laps, cold shuts or any other malformation or defect impairing the intended use or serviceability.

4.5 Sampling procedures.

4.5.1 *Lot size.* All complete production units of the same size, produced under the same manufacturing processes and conditions, at the same plant, and which are scheduled for delivery to the Government at one time, shall be divided into lots of not more than 100 units.

4.5.2 *Samples.* Random samples of production units shall be selected from each lot for testing. The selection, handling, and disposition of samples shall be according to the provisions of MIL–STD–105 at inspection level L–7. Samples shall be tested as specified in 4.6.3. and 4.6.4.

4.6 Tests.

4.6.1 *Proof test.* Each complete block shall be reeved, without using becket, with wire rope of the size recommended for use thereon. The proof test load shall be suspended from the hook for a period of not less than ten minutes. The block shall then be reeved with the wire rope dead ended at the becket and the load reapplied for ten minutes.

4.6.2 *Hooks.* Each hook shall be subjected to a magnetic particle test in accordance with MIL–STD–271 and shall conform to 3.6 prior to assembly.

4.6.3 *Overload test.* The block shall be reeved, without using becket, and a load

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equal to 1-½ times the safe working load suspended from the hook. The load shall then be raised and lowered a minimum of 20 feet at speed selected by manufacturer.

4.6.4 *Service test.* With the block reeved as in 4.6.3, a load equal to the safe working load shall be raised and lowered a minimum of 20 feet. The cycle shall be repeated a minimum of 10 times at speeds approximating those encountered in crane service. The line speed shall be not less than 120 feet per minute. The load shall be rotated through 360 degrees a minimum of three times to determine conformance to 3.7.2. With load removed, the block shall be raised and lowered three times to determine conformance to free fall requirements specified in 3.4.

4.7 **Acceptable quality level.** Any complete block or component thereof failing to pass any test specified herein shall be considered defective. The Acceptable Quality Level for samples shall be 2.5 percent defective.

4.8 **Preparation for delivery inspection.** The preservation, packaging, packing and marking of the blocks shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

(**Civil agencies.** Fed. Std. No. 102 should be referred to for definitions and application of the various packaging levels for supplies and equipment.)

5.1 **Preservation and packaging.** The preservation and packaging shall be level A or C as specified.

5.1.1 *Level A.* The blocks shall be preserved and packaged in accordance with MIL–P–3865.

5.1.2 *Level C.* The blocks shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 **Packing.** The packing shall be level A, B, or C, as specified.

5.2.1 *Levels A and B.* The blocks shall be packed in accordance with MIL–P–3865.

5.2.2 *Level C.* The blocks shall be packed or prepared for shipment in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to the carriers at lowest rates.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking specified in the invitation to bid, marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* Marking for shipment shall be in accordance with MIL–STD–129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and data of this specification.
(*b*) Size (see 1.2).
(*c*) Wire rope size for sheaves (see 3.5).
(*d*) When non-safety type hook is required (see 3.6).
(*e*) When material designation and heat-treatment markings are required (see 3.6).
(*f*) Length of time inspection records will be kept (see 4.1).
(*g*) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(*h*) Marking, if different (see 5.3).

6.2 **Transportation description.** The applicable transportation description is:

Tackle blocks, snatch blocks or sheaves, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.
Truckload minimum weight 24,000 pounds.

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